

- Some of the homework assignments will require you to run programs or modify certain scripts. While you can choose the programming language that you would like to use I will post most of the scripts in MATLAB. As a student you have access to matlab through the university. Matlab Online is a good way to test it without having to install it on your computer. You can create an account with your UT email and have access to it. [MATLAB Online - MATLAB & Simulink](https://github.com/NelsonUpenn/PMLS-MATLAB-Guide/blob/master/MLguide1-3.pdf) Optional: If you are not familiar with programming or with MATLAB, you can complete the “Guide” tutorial to help you familiarize yourself with the program. An excellent guide can be found here: <https://github.com/NelsonUpenn/PMLS-MATLAB-Guide/blob/master/MLguide1-3.pdf>

- (10 pts) Consider a set of 3 six-sided dies, with the faces labeled 1 through 6 on each die.
 - Calculate the entropy of the system.

The total entropy of the system is ~ 5.38 when there are 6^3 microstates in the system.

- Calculate the average score after a large number of rolls.

The average score should be 3.5 for each dice. Thus, the average total score should be 10.5.

- Consider, instead a die with 18 sides labeled 1 through 18, is the average score the same as part b? Why or why not?

The average score for the 18-sided die is 9.5. When rolling 3 separate 6-sided die, the average score is the sum of independent events, while the 18-sided die is one event. This creates different distributions and therefore different averages.

- Write a computer program to list all possible outcomes of the three six-sided dice (note consider the dice are not interchangeable so 1,2,3 is not the same state as 1,3,2 etc).

```
roll = [(d1,d2,d3) for d1 in range(1,7)
        for d2 in range(1,7)
        for d3 in range(1,7)]
```

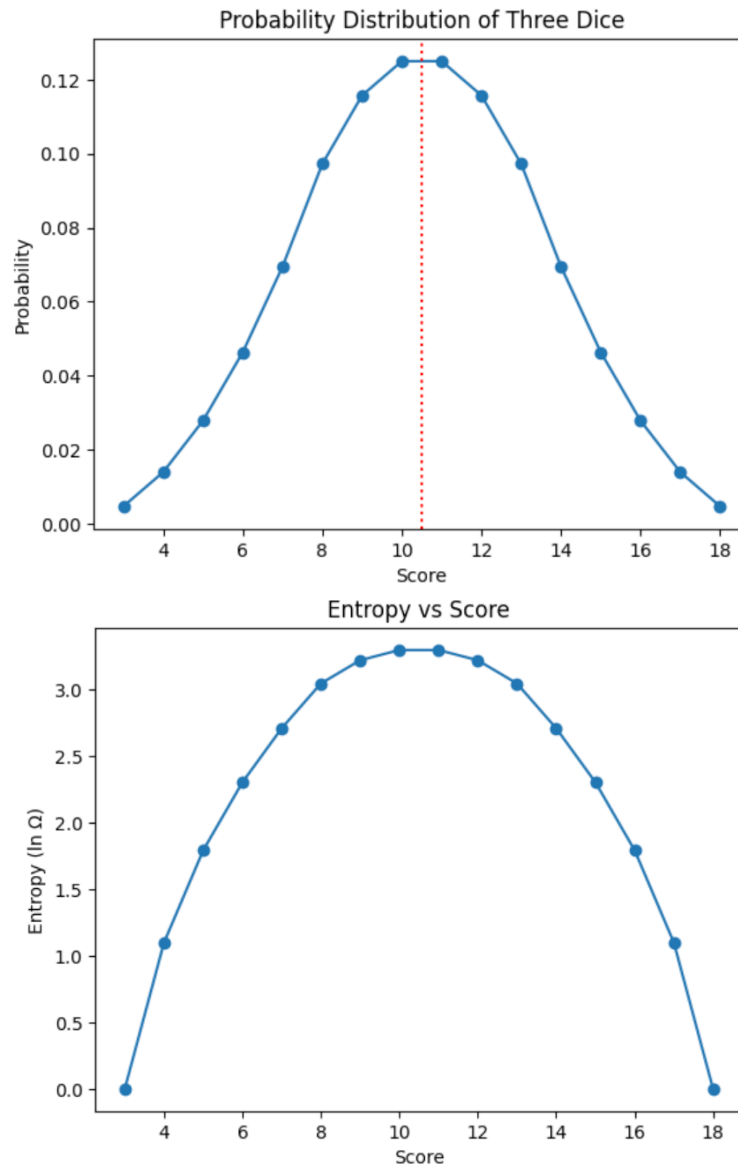
- The minimum score for this system is $1+1+1=3$, the maximum is $6+6+6=18$; Extend your computer program to compute the probability of finding each individual score between 3 and 18. Make a plot of the probability and entropy of each score.

```
[12]: import numpy as np
      import matplotlib.pyplot as plt

[13]: # initialize the rolling and the sum for each of the 3 rolls
      roll = [(d1,d2,d3) for d1 in range(1,7)
              for d2 in range(1,7)
              for d3 in range(1,7)]
      sums = [sum(r) for r in roll]

[14]: # consider all possible scores 3-18
      # counts scores number of microstates that produce each total score s
      scores = np.arange(3,19)
      counts = np.array([sums.count(s) for s in scores])

[15]: # Convert multiplicity to probability
      prob = counts / 216
      # Microstate entropy per score
      entropy = np.log(counts)
```



3. (10 pts) In a completely random set of numbers, all digits (1 through 9) should appear with equal frequency. However, in the real world, the distribution of leading-digits is non-uniform, with lower digits appearing more frequently. For example, a 1 is more likely to appear as a first digit than a 2, and so on. This is true across many data sets including scientific data and financial data, and the phenomenon is known as Benford's law. The law states that the real-world probability of having any number 1-9 as the leading digit is given by the following equation:

$$P_n = \log_{10} \left(\frac{n+1}{n} \right); n = 1, 2, 3, \dots, 9$$

Calculate the loss in entropy of the distribution predicted by Benford's law compared to a purely random set of numbers where each digit appears with the same frequency.

Using the Shannon definition of entropy, we find that for a uniform distribution of digits, the total entropy is ~ 2.197 . Using the probabilities given by Benford's law, we find that the true entropy for the distribution of digits is ~ 1.99 . Thus, the loss in entropy given by Benford's law is ~ 0.2 .

4. (10 pts) The average score obtained from rolling a six-sided die many times is $(1+2+3+4+5+6)/6 = 3.5$. You just got a job at an online gambling website, and your boss asked you to program a "digital die" that gives the house an advantage by producing an average score of 3 instead of the unbiased score of 3.5. You want to produce the distribution that is closest to a uniform distribution so that the players do not suspect that the game is rigged. Calculate the probability of rolling each number 1...6 in this distribution.

Given the constraint

$$\langle \varepsilon \rangle = 3$$

$$3 = \frac{\sum i e^{-\beta \varepsilon_i}}{\sum e^{-\beta \varepsilon_i}}, \therefore \beta = 0.371$$

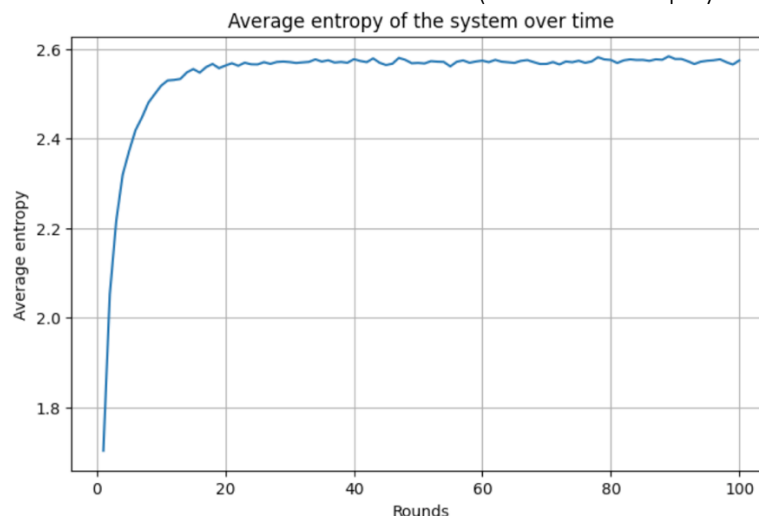
Using

$$P_i = \frac{e^{-\beta \varepsilon_i}}{Q}$$

We find the probabilities:

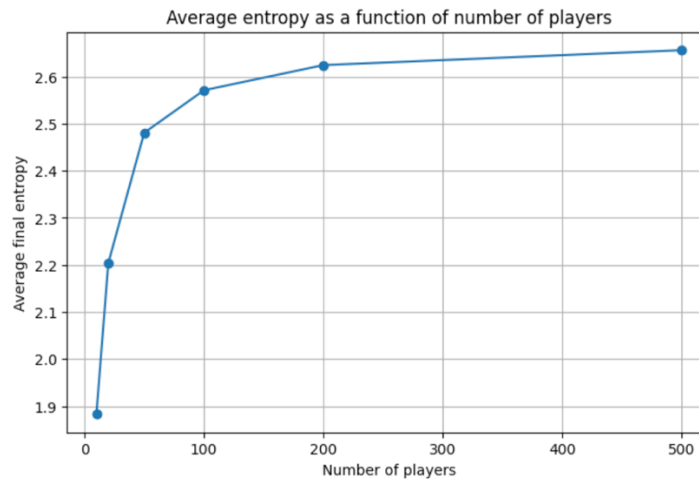
Score	P_i
1	0.288
2	0.199
3	0.138
4	0.096
5	0.067
6	0.046

5. (20 pts) Using the code that simulates the "Boltzmann Game" (available on Canvas), plot the average entropy of the system:
- as a function of the number of "rounds" (each individual plays one game per round).



At each round, the probability distribution of money among the players was calculated along with the Shannon entropy. The results were averaged over several simulations.

- as a function of the number of players (from 10 to 1000).



The same simulation was run for different numbers of players while keeping the number of rounds fixed. The final entropy for each system was computed and averaged over simulations. Then entropy versus the number of players was plotted as shown.

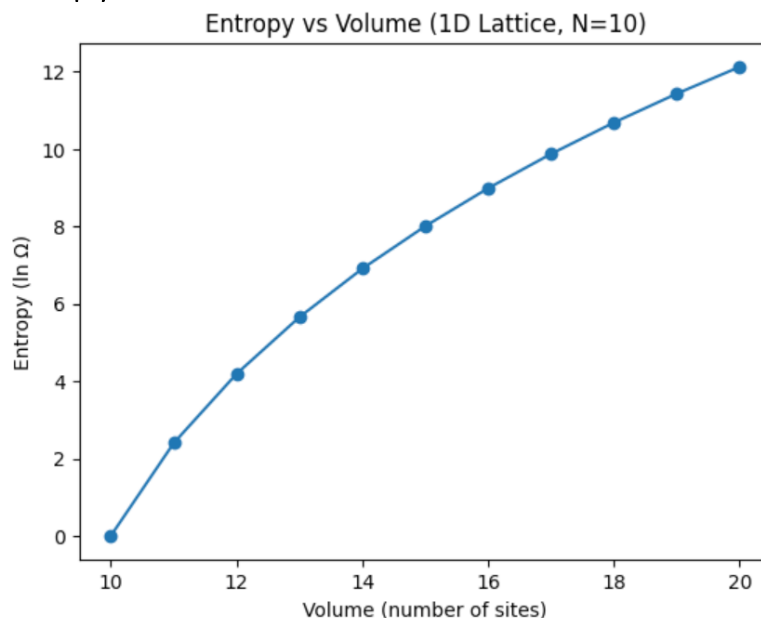
6. (20 pts) Consider an ideal gas in a 1D lattice model, similar to the example we discussed in lecture (Dill 2.2). The total number of gas molecules is 10;

- a. Calculate the entropy of the system if these 10 molecules are distributed over a volume of 10, 11, 12, 13 ... 20 sites. Make a plot of the entropy as a function of the volume.

To calculate the number of ways to choose which sites will be occupied, we can use

$$\Omega = \frac{V!}{N!(V-N)!}$$

Then, using $S = \ln(\Omega)$, we find that entropy increases with the volume but begins to flatten out as when $V \gg N$, most sites are empty.



- b. Consider an ideal gas. Compute the change in entropy by isothermally doubling the volume of the container from an initial volume V_1 to a final volume $V_2 = 2 \cdot V_1$. Make a plot of the increase in entropy as a function of volume and compare the shape of the plot with the lattice model in part a.

For an isothermal doubling of the volume, the entropy increases by $\ln(2)$ per molecule. Unlike the lattice gas model, where the entropy gradually rises and then begins to flatten as volume increases, the entropy increase for the ideal gas is constant and does not show a plateau as each expansion always provides the same relative increase in accessible microstates.

7. (10 pts) In the “money game” that we played in lecture, the only constraint is that the total amount of money in the room was constant, which produced an exponential distribution that is somewhat similar to the actual distribution of wealth in the United States. Suppose you want to make an argument for a “equality-focused” society where no individual has more than 5 times the amount of money compared to the average individuals. Using the Matlab code provided, perform a simulation where you give each participant 10 coins and exclude participants that already have 50 coins from receiving more coins. Does this affect the distribution? Why or why not?

The distribution changes when we cap coins at 50. In the original game, the distribution was exponential, allowing a few players to accumulate many coins. Adding the cap reduces the number of accessible microstates, preventing extreme wealth and producing a more equal distribution where no one can hoard all the coins.

8. (20 pts) In 1773 Benjamin Franklin performed a famous experiment where he dropped a tablespoon of olive oil into a pond and saw that the oil spread over a wide area over the surface of the water. While Franklin did not pursue this experiment scientifically, by measuring the approximate area of the oil patch, other scientists were able to estimate the thickness of the oil film over the water and were able to deduce the approximate size of an oil molecule with the assumption that the layer was only one molecule thick. For a compelling account of these experiments read: *Ben Franklin Stilled the Waves: An Informal History of Pouring Oil on Water with Reflections on the Ups and Downs of Scientific Life in General* by Charles Tanford.

From an entropic point of view, the number of microstates of the oil molecules on the surface of the water is fewer compared to the microstates of liquid oil as the molecules form a monolayer on the surface. Explain why this process is thermodynamically favorable. In other words, why doesn't the oil stay as a single "drop" on top the water? Hint: consider all the interactions that are present in the system (oil, water, air) before and after the oil is dropped on the water.

This process is thermodynamically favorable because of the interactions between oil, water, and air. Oil spreads on the water surface to minimize high-energy interactions with the polar water molecules. Although forming a monolayer reduces the entropy of the oil molecules, the overall enthalpy of the system decreases due to the replacement of high-energy oil-air and water-air interfaces with lower-energy oil-water interfaces. As a result, the total free energy of the system decreases, making the spreading process spontaneous